Nico Carlsen Daheim né Daheim

nico.daheim@rwth-aachen.de • ndaheim • ndaheim • ndaheim_
~ 1,200 citations, H-index: 16

Research Interests

My research has focused on efficient methods for improving LLMs by using uncertainty and tutoring applications. I have published novel approaches to problems ranging from pretraining, model merging, and post-training to test-time scaling. Currently, I am excited about improving data mixes, pretraining stability, post-training, and LLM efficiency.

Education

ELLIS PhD in Computer Science (summa cum laude), TU Darmstadt 09/2022–07/2026
Supervised by Prof. Dr. Iryna Gurevych, Prof. Dr. Mrinmaya Sachan (ETH Zurich).
Thesis: Practical Uncertainty for Language Models - handed in 07/2026, defended 09/2026, final certificate pending.

M.Sc. Data Science (cum laude), RWTH Aachen University 10/2019–08/2022

B.Sc. Computer Science (Minor: Mathematics), University of Koblenz 04/2016–03/2019

Experience

Postdoctoral Researcher, TU Darmstadt 07/2026–ongoing
Continuing to work with Prof. Mohammad Emtiyaz Khan on scaling variational learning.

Visiting Researcher, ETH Zurich 11/2024–08/2025
Visiting Prof. Dr. Mrinmaya Sachan's LRE Lab to work on LLMs for tutoring applications.

Student Research Assistant, RWTH Aachen University 07/2020–07/2022
Research on document-grounded dialog systems at Prof. Hermann Ney's lab.

Intern & Data Scientist, Deutsche Telekom 04/2019–03/2020
Dialog systems for customer service, semantic search, text analytics. Gap due to role transition: 10/2019.

Student Research Assistant, University of Koblenz 03/2018–03/2019
Document clustering and sentiment analysis for social data.

Selected Publications

Parameter Exploration for RLVR via Variational Learning — NeurIPS 2026
V. Venkatkrishna, **N. Daheim**, I. Gurevych

SOAP-Bubbles: Structured Weight Uncertainty for Neural Networks — NeurIPS 2026
AR Minut, **N. Daheim**, M. Miani, ME Khan, W. Lin, T. Möllenhoff

SVRG and Beyond via Posterior Correction (Oral) — ICML 2026
N. Daheim, T. Möllenhoff, ML Ang, ME Khan

Aligning LLMs with Pedagogy using Reinforcement Learning — EMNLP 2025
D. Dinucu-Jianu*, J. Macina*, **N. Daheim**, I. Hakimi, I. Gurevych, M. Sachan

MathTutorBench: A Benchmark for Measuring Open-ended Pedagogical Capabilities of LLM Tutors — EMNLP 2025
J. Macina, **N.Daheim**, I. Hakimi, M. Kapur, I. Gurevych, M. Sachan

Uncertainty-Aware Decoding with Minimum Bayes Risk — ICLR 2025
N. Daheim, C. Meister, T. Möllenhoff, I. Gurevych

Stepwise Verification and Remediation of Student Reasoning Errors with Large Language Model Tutors — EMNLP 2024
N. Daheim*, J. Macina*, M. Kapur, I. Gurevych, M. Sachan

Variational Learning is Effective for Large Deep Networks (Spotlight) — ICML 2024
Y. Shen*, **N. Daheim***, B. Cong, P. Nickl, GM Marconi, C. Bazan, R. Yokota, I. Gurevych, D. Cremers, ME Khan, T. Möllenhoff

Model Merging by Uncertainty-Based Gradient Matching — ICLR 2024
N. Daheim, T. Möllenhoff, EM Ponti, I. Gurevych, ME Khan

MathDial: A Dialogue Tutoring Dataset with Rich Pedagogical Properties — EMNLP Findings 2023
J. Macina*, **N. Daheim***, SP Chowdhury*, T. Sinha, M. Kapur, I. Gurevych, M. Sachan

Contributions to Book Chapters

Handbook on Bayesian Deep Learning - Natural-gradient-based Variational Inference, Bayesian LLMs
BayesAI Consortium 2026

Invited Talks

Variational Learning for LLMs - Bayes in Modern AI Workshop 2026
SVRG and Beyond via Posterior Correction - RIKEN AIP 2026
Practical Bayesian Uncertainty for LLMs - INSAIT 2025
Model Merging by Uncertainty-Based Gradient Matching - Flatiron Institute 2025
Model Merging by Uncertainty-Based Gradient Matching - RIKEN AIP 2024
Adapting Language Models by Model Merging - MBZUAI 2023
Adapting Language Models by Model Merging - University of Edinburgh 2023
Language Models & Model Merging - RIKEN AIP 2023

Tutorials

LLMs for Education - Building Educational Applications (BEA) Workshop, ACL 2025

Teaching Experience

Generative AI Seminar - TU Darmstadt 2023–2024
Machine Learning - Teaching Assistant, RWTH Aachen University 2021–2022

Reviewing Service

ARR 12/23–08/26, ICML '24-'26 (Gold Reviewer), ICLR '25-'27, NeurIPS '25-'26 (Top Reviewer), UncertaiNLP 24 – 26, QUESTION@ICLR '25 , UniReps@NeurIPS '24, Doc2Dial@ACL '23

Awards

Springorium Medal - RWTH Aachen University 2023
Porsche IT Campus RWTH Aachen Scholarship - RWTH Aachen University 2020

Skills

Programming: Python, PyTorch, huggingface
Languages: German (native), English (near native)